A set of matchings closed under Knuth equivalence is Schur-positive with two-row expansion

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Editorial note. The counted statement is the author's labelled corollary 'cor:closed under knuth like is schur positive' (source lines 1214-1221); the proof environment begins on the next line after \end{corollary} apart from one blank line, so the pairing is by adjacency.

Editorial note. Two context spans are retained: the subsection label invoked by the proof and the theorem characterising Knuth equivalence, whose statement is retained so that the invoked label is defined inside the package.

Editorial note. The retained preamble is the editorially prepared standalone preamble retained for this source.

Editorial note. One external reference records the Schur-positivity criterion invoked at the last line; it is a navigation pointer only.

Editorial scope. One original result of the pinned publisher article, counted once: Schur-positivity of sets closed under Knuth equivalence (author label cor:closed under knuth like is schur positive).

Allowed standard prerequisites (editorial).

- Knuth equivalence for matchings. Two matchings are Knuth equivalent if they can be connected by the insertion and deletion moves of the section.. The hypothesis is that the set is closed under this equivalence.
- Short statistic and unstable chords. Short is the statistic recording the positions of the short chords, and k denotes the number of unstable chords of a matching.. The expansion is written in terms of s_{N-k,k} with k read off from any matching in the class.
- The transformation F. F(m) = (core(m), T(m)) is the bijection from matchings to pairs consisting of a core and a standard Young tableau, discussed in the retained subsection.. Its restriction to a Knuth class is the statistic-preserving bijection at the heart of the proof.

Source reference. Original theorem 2.5: sources/alco-351-matching-schur-2024/publisher-source.tex, source lines 368–368. Original criterion of Adin and Roichman invoked at source line 1228. It is a navigation pointer: the counted corollary's own proof is retained line for line and nothing is replaced by this reference. The reference number was inferred by counting the theorem-class environments of that section of the pinned source; it is not confirmed by the retained bibliography or by a label-number dump of the original, so it must be read as a navigation pointer only.

Theorem 0.1. Two matchings $m_1, m_2 \in \mathcal{M}_N$ are Knuth equivalent if and only if $\mathrm{core}(m_1) = \mathrm{core}(m_2)$.

Corollary 0.2. If a set $\mathcal{M} \subseteq \mathcal{M}_N$ is closed under Knuth equivalence then it is Schur-positive with respect to Short. Moreover, if $\mathcal{M}$ is a Knuth equivalence class then its generating function is

$$\mathcal{Q}(\mathcal{M}) = s_{N-k,k},$$

where k is the number of unstable chords of some arbitrary matching $m \in \mathcal{M}$.

Proof. Clearly, a disjoint union of Schur-positive sets is Schur-positive. Therefore, it suffices to establish the statement for equivalence classes. Let $\mathcal{M} \subseteq \mathcal{M}_N$ be an equivalence class. By Theorem 0.1 there exist $0 \leq k \leq \frac{N}{2}$, $f \geq 0$ and $m_0 \in \mathcal{M}_{N-2k,f}(\mathrm{Short} = \emptyset)$, such that $\mathcal{M} = \{m \in \mathcal{M}_{N,f} \mid \mathrm{core}(m) = m_0\}$. Recall the transformation $F : m \mapsto (\mathrm{core}(m), T(m))$ discussed in Section , and consider its restriction to $\mathcal{M}$. We obtain that the restricted transformation

$$F\big|_{\mathcal{M}} : \mathcal{M} \to \{m_0\} \times \mathrm{SYT}(N-k, k)$$

is a statistic-preserving bijection. Applying Theorem 2.5 completes the proof. □

References

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